

MOVIEREC – MOVIE RECOMMENDATION SYSTEM

AIM

The major objective of this project is to create a smart Movie Recommendation System that uses the The Movie Database (TMDb) API to recommend films to users based on their tastes. The system integrates genre-based filtering, user ratings, and popularity ratings to provide a hassle-free movie discovery experience. Through the use of contemporary web technologies such as React and Material-UI, the project seeks to provide a responsive, user-friendly interface that appeals to both casual viewers and film enthusiasts looking for personalized recommendations.

OBJECTIVES

1. User-Centric Interface Development

- Design an intuitive UI with search functionality and genre selection filters
- Implement responsive layouts using Material-UI components for cross-device compatibility
- Ensure accessibility standards (WCAG) for color contrast and keyboard navigation

2. Intelligent Recommendation Engine

- Integrate TMDb API to fetch real-time movie data (popular/search/discover endpoints)
- Develop algorithm to suggest movies based on:

o User-selected genres (multi-select filter) o

Community ratings (TMDb vote averages) o

Personalized ratings (localStorage persistence)

- Implement debounced search to optimize API calls

3. Personalized Rating System

- Create 5-star rating interface with MUI Select component
- Store user ratings in localStorage for persistent preferences
- Visual feedback using filled/empty Star icons

4. Performance Optimization

- Implement lazy loading for movie posters
- Cache API responses using browser localStorage
- Reduce re-renders with React memoization

5. Error Handling & User Feedback

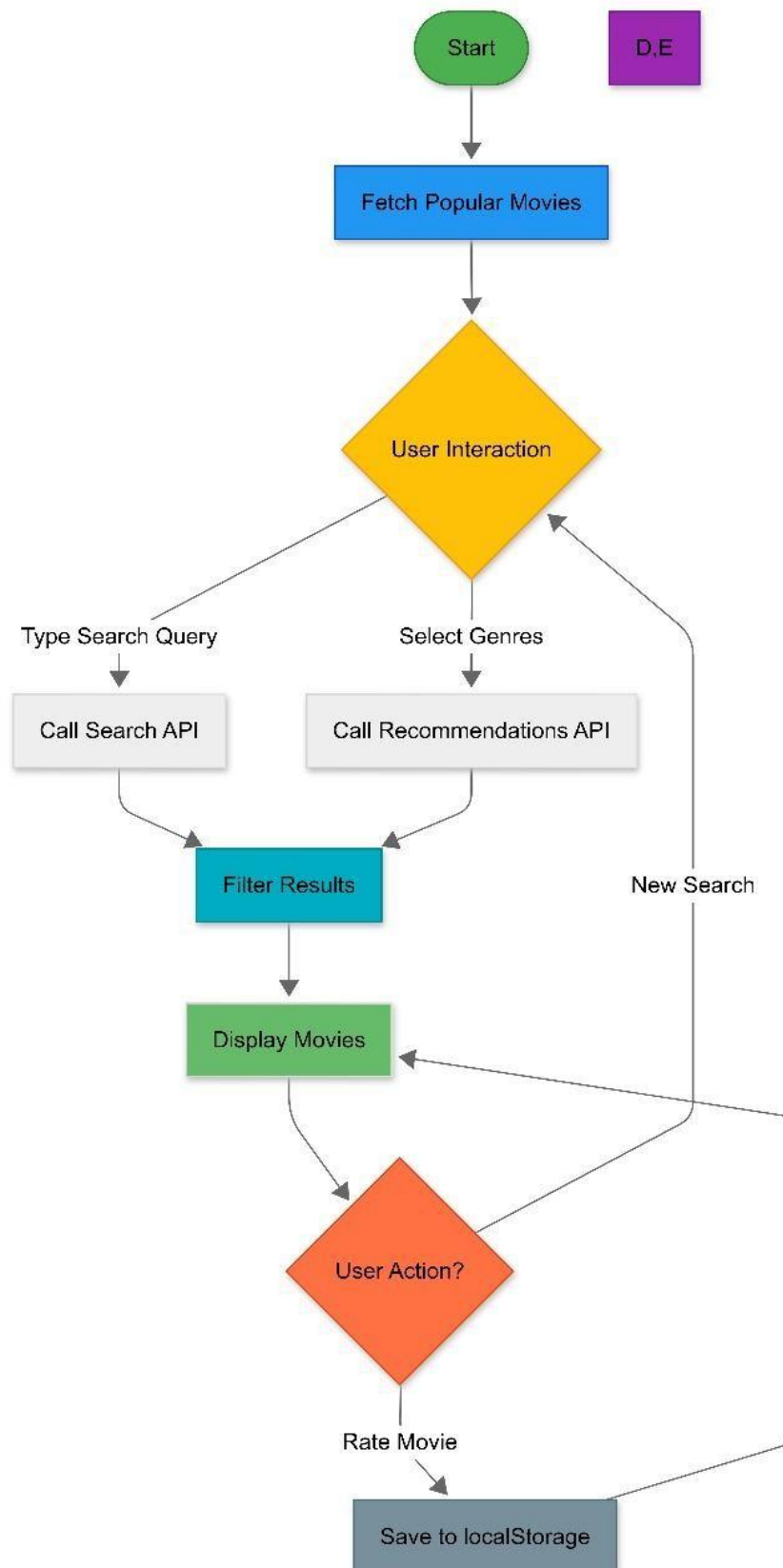
- Display loading states with Circular Progress indicators
- Handle API errors with user-friendly messages
- Implement empty state UI for no-results scenarios

6. Documentation & Scalability

- Create system flowchart (Mermaid.js) for technical documentation
- Structure code for future enhancements:

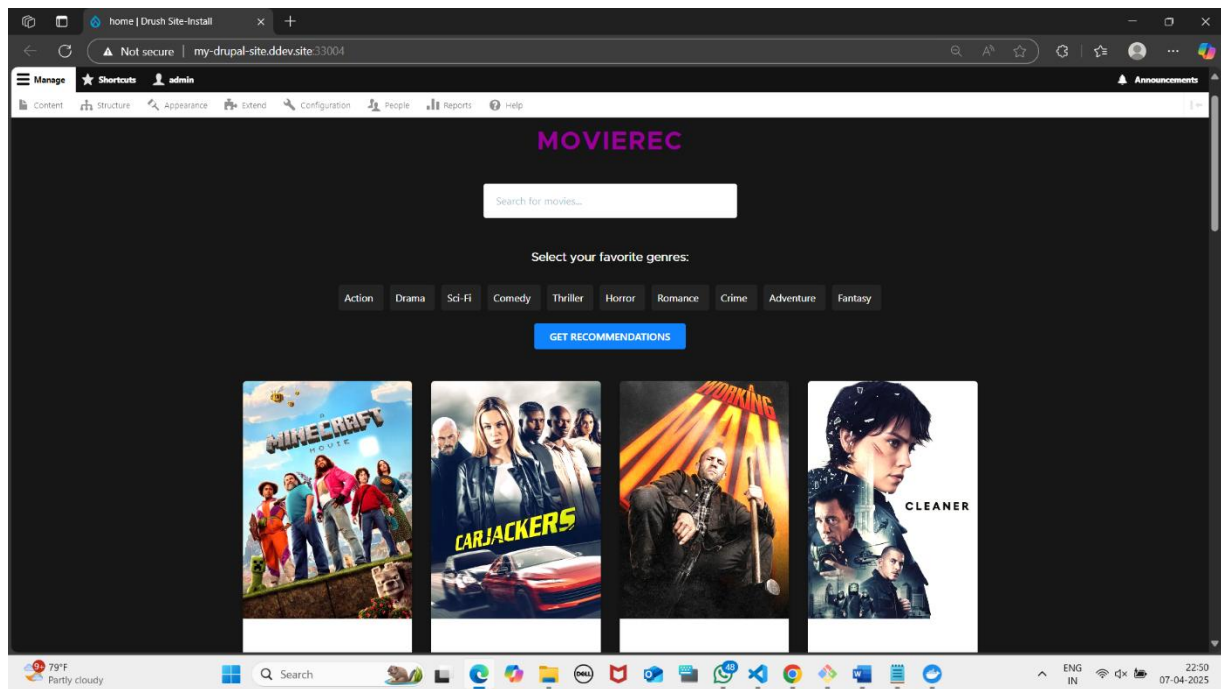
- o User authentication
- o Advanced filters (release year, actor)
- o Machine learning recommendation

FLOWCHART:



FUNCTIONALITIES:

WEBSITE OVERVIEW:



- This is the website overview of our mini project WEBREC , a movie recommendation system .

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1.Fetch Popular Movies

Uses a reliable movie database API TMDb to fetch trending/popular movies.

- Implements caching to store recently fetched movies and reduce repeated API calls.
- Displays movie posters, titles, and ratings in a visually appealing format.

2.Get movie recommendations based on categories

Allows users to select genres (e.g., action, comedy, thriller) for personalized recommendations.

- Utilizes API filters to fetch movies based on selected categories dynamically.



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Select your favorite genres:

ActionDramaSci-FiComedyThrillerHorrorRomanceCrimeAdventureFantasy

GET RECOMMENDATIONS



Arena Wars

5.12024

Rate this



Kingsman: The Golden Circle

7.02017

Rate this



The Tuxedo

5.72002

Rate this



The Avengers

4.41998

Rate this

73°FPartly cloudy

Search



ENGIN

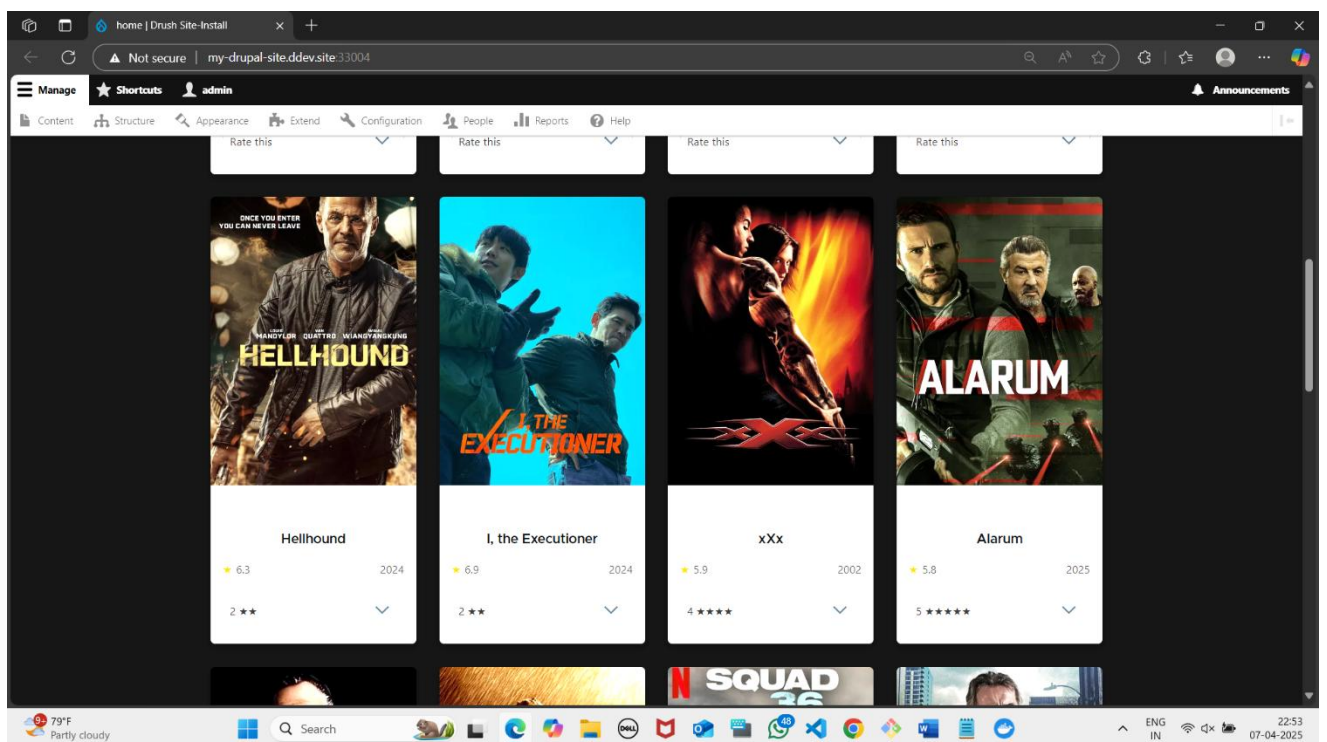
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3. Rating System for Movies with local storage:

Enables users to rate movies and store ratings in local storage for persistence.

- Displays average user ratings alongside fetched ratings from the movie database.
- Allows users to edit or remove their ratings, updating the UI dynamically.

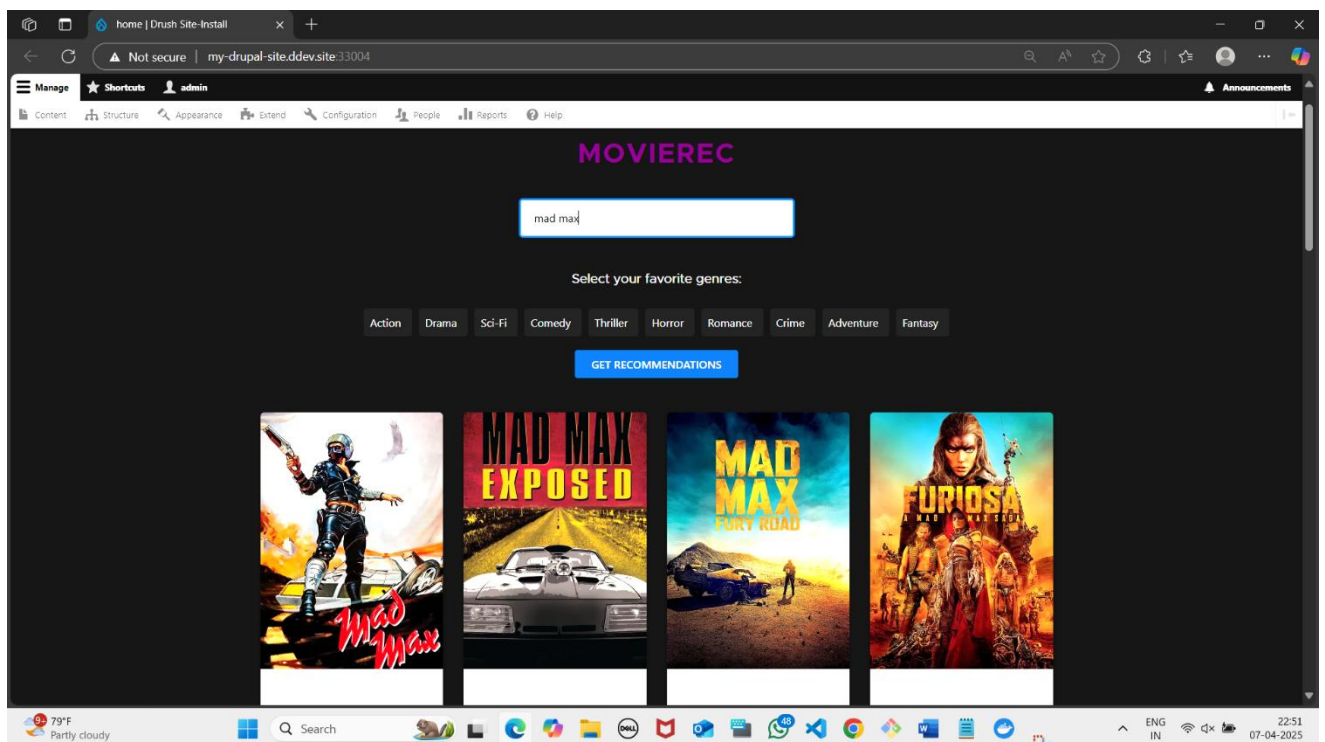


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4.Auto-Complete feature in the searchbar

Implement a debounce function to reduce API calls while typing.

- Fetch and display movie suggestions dynamically as users type.
- Support keyboard navigation (up/down arrows) to select suggestions easily.



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